FreePBX Care
Standalone server guide

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Run every command directly ON THE PBX SERVER as root, over SSH or its console. This tool configures monitoring and outgoing email for an existing FreePBX installation. It does not install Debian, FreePBX or extensions.

1. BEFORE STARTING

Supported platform: FreePBX 17 on Debian 12, with systemd and Python 3. This watchdog expects LOCAL MariaDB with root-accessible socket ping. Remote database deployments require a separately reviewed monitoring policy. Have a current, recoverable PBX backup and the SMTP mailbox password from the company's secure credential store. Confirm the correct customer/server. Use an approved maintenance window for migration or outgoing-email changes. Postfix SMTP changes also affect other outgoing mail, including voicemail.

Become root if necessary:

```
sudo -i
```

Check the platform and clock:

```
cat /etc/os-release
fwconsole --version
command -v python3
timedatectl status
```

If the platform differs, Python is missing, or the clock is wrong, have the administrator resolve that first. Do not bypass platform checks. Working DNS and access to GitHub and the SMTP relay are required.

Choose the correct path:

```
NEW server, no legacy monitors: section 2, then section 4.
OLD server with legacy monitors: sections 2 and 3, then section 4.
Existing working FreePBX Care installation: section 7; no legacy cleanup.
Only changing email settings: section 5, then section 6; no cleanup needed.
```

2. DOWNLOAD THE CURRENT SCRIPTS

Download the main script. Copy the full command, including the && lines:

```
curl -fSL --retry 3 \
  https://raw.githubusercontent.com/workimirp201/freepbx-script/refs/heads/main/freepbx-care.sh \
  -o /root/freepbx-care.download.sh &&
bash -n /root/freepbx-care.download.sh &&
install -m 700 /root/freepbx-care.download.sh /root/freepbx-care.sh
```

Check the version:

```
bash /root/freepbx-care.sh --version
```

This guide matches 2026.10.27-module-review-r6. Use matching instructions if a later release changes the prompts. If a download or syntax check fails, stop and resolve the error before proceeding.

For OLD servers only, also download the standalone cleaner:

```
curl -fSL --retry 3 \
  https://raw.githubusercontent.com/workimirp201/freepbx-script/refs/heads/main/cleanup-legacy-freepbx.sh \
  -o /root/cleanup-legacy-freepbx.download.sh &&
bash -n /root/cleanup-legacy-freepbx.download.sh &&
install -m 700 /root/cleanup-legacy-freepbx.download.sh /root/cleanup-legacy-freepbx.sh
```

Run these tools with bash, even though they contain embedded Python.

3. OLD SERVERS: CLEAN UP BEFORE INSTALLING THE REPLACEMENT

Skip this section for a new server with no legacy monitors. Before cleanup, record the old property names and selected extension numbers for manual entry in option 10. Do not assume the replacement imports them.

A. NORMAL MIGRATION: KEEP EXISTING SMTP DURING CLEANUP

First preview; this does not apply cleanup:

```
bash /root/cleanup-legacy-freepbx.sh
```

Review the listed files and cron jobs. Then apply:

```
bash /root/cleanup-legacy-freepbx.sh --apply
```

Wait for 'Cleanup complete' or 'No legacy changes needed'. For errors, use section 8; do not proceed as if cleanup succeeded.

The cleaner removes recognized root cron entries and archives these monitors:

```
/usr/local/bin/asterisk-monitor.sh
/usr/local/bin/extension_monitor.sh
```

It archives exact recognized installer copies, preserves modified/unknown installers, and disables a recognized existing FreePBX Care timer. It does not restart Asterisk. SMTP settings and sender rewrites stay by default. Backups, including root crontab and relevant Postfix files, are under:

```
/var/lib/freepbx-care/legacy-backup-TIMESTAMP/
```

Other users' cron jobs, custom paths and schedulers need administrator review. Do not clear the entire crontab. Rejected chained commands must be reviewed and separated from unrelated tasks by the administrator. Continue promptly to section 4 because legacy monitoring is now disabled.

B. OPTIONAL: RESET OLD SMTP SETTINGS TOO

Use only when deliberately replacing the old relay settings and prepared to complete new SMTP setup immediately. This is not required for normal migration. Use this path instead of A when a reset is planned:

```
bash /root/cleanup-legacy-freepbx.sh --reset-smtp
bash /root/cleanup-legacy-freepbx.sh --apply --reset-smtp
```

A recognized reset archives old relay password files, detaches recognized old sender rules, and defers smtp/relay delivery until current SMTP setup completes. Queued messages are retained subject to normal queue expiry. Unknown/custom

relay configurations are not automatically reset; read the reported plan. Continue section 4 INCLUDING option 3. Do not remove the reset marker manually, bypass mail deferral, or delete/flush the queue to fix a pending reset.

The separate --remove-sender-rewrites flag is for an administrator deliberately changing recognized old sender rules. It can change voicemail sender identity; it is not part of routine cleanup.

4. SETUP FOR NEW OR CLEANED OLD SERVERS

Open the menu:

```
bash /root/freepbx-care.sh
```

Menu options:

```
1 Configure Asterisk monitor
2 Install/update monitoring
3 Configure SMTP relay
4 Read-only health/status
5 Send test email
6 Pause monitoring
7 Resume monitoring
8 Disable monitoring
9 Collect diagnostics (no restart)
10 Configure extension monitoring
11 Read-only extension status
12 Configure MariaDB / CRM checks
13 Review/disable unused optional modules
14 Investigate crash/resource issues
0 Exit
```

Recommended order: 1 -> 3 -> 5 -> 4 -> 2 -> 0. When extension alerts are required, use 1 -> 3 -> 5 -> 10 -> 11 -> 4 -> 2 -> 0. Review option 12 before installation: for this fleet use MariaDB y, CRM n. Optional module review is option 13; Enter at its selection prompt audits only. Wait for actual test-email receipt before enabling monitoring with option 2. If intentionally keeping already verified SMTP on an old server, skip option 3 but still do option 5 and confirm receipt. After an SMTP reset, option 3 is mandatory.

OPTION 1 - CONFIGURE ASTERISK MONITOR

Enter these company production values. Type each explicitly if the displayed default differs:

```
Alert recipient:
  monitor@famecomputers.com
Sender email (blank keeps system default):
  support@famepbx.com
Enable automatic recovery? y/n:
  y
Defer restart while CLI reports active channels? y/n:
  y
Consecutive failed checks:
  3
Restart cooldown seconds:
  1800
Maximum restart attempts per hour:
  2
AMI settings: auto or manual?:
  auto
```

For an observation-only pilot choose n for automatic recovery. Alerts and health monitoring still work; change it to y with option 1 after validation. Change sender/recipient only if approved for this server/customer.

AMI 'auto' reads this server's own local FreePBX credentials. It is separate from SMTP: never use the email password as an AMI secret. If auto fails, collect status and ask the administrator instead of guessing manual credentials.

OPTION 3 - CONFIGURE SMTP

Use the exact answers in section 5.

OPTION 5 - SEND TEST EMAIL

'Accepted by local sendmail' means accepted into the local queue only. Confirm receipt at monitor@famecomputers.com. If missing, follow section 6.

OPTION 4 - READ HEALTH/STATUS

Expect CLI true, AMI healthy and healthy true. A healthy result may show restartable false: a healthy Asterisk does not need recovery. Fix CLI/AMI errors before installing monitoring. Check for unintended pauses too. The database field must also show healthy true. A stopped database must be started and checked manually before a first installation can pass preflight.

OPTION 2 - INSTALL/UPDATE MONITORING

Installs and enables the systemd timer. Configuration alone does not schedule checks. Resolve installation errors before handoff. Installation does not restart Asterisk, but enabled automatic monitoring may recover it later.

OPTION 0 - EXIT, THEN VERIFY

```
systemctl is-enabled freepbx-care.timer
systemctl is-active freepbx-care.timer
systemctl list-timers --all freepbx-care.timer
bash /root/freepbx-care.sh status
```

Expected timer results: enabled and active. The oneshot service may be inactive between checks; inspect its journal for actual failures:

```
journalctl -u freepbx-care.service -n 50 --no-pager
```

Test normal incoming/outgoing calls and a voicemail email. Record hostname, script version, health, timer status, test-mail receipt, voicemail result and whether automatic recovery is enabled. Never record passwords.

OPTION 10 - MANUALLY CONFIGURE EXTENSION MONITORING

Extension monitoring is OPTIONAL and OFF until a technician configures it. Use PJSIP extension/AOR numbers from THIS PBX, not numbers from another property. No groups are automatically imported from the old installer. If migrating, record the old property names and extension lists before cleanup, or review the archived old script with the administrator. Do not execute it again.

Choose option 10, or run:

```
bash /root/freepbx-care.sh extensions
```

Example answers (replace the example property and numbers with the real ones):

```
Enable extension monitoring? y/n [n]:
  y
Extension consecutive failed checks [3]:
  3
Replace property/extension groups? y/n [y]:
  y
How many properties/groups [1]:
  1 (or the actual number of properties/groups on this server)
Property/group 1 name:
  Red Roof Front Desk (example; enter the actual site/group name)
PJSIP extension numbers (spaces or commas):
  101 102 103 (example; enter the actual selected numbers)
Outage mode: all or any [all]:
  all
```

Repeat name/numbers/mode for every group when configuring multiple properties. Use 'all' for the old behavior: alert only when ALL selected extensions are unavailable, recover when AT LEAST ONE is available again. Recovery does not mean every extension is back. This is recommended for detecting a property outage. Use 'any' when EVERY selected phone is critical: alert if any selected extension is unavailable, recover when all are available. To get independent alerts for individual phones, create a separate single-extension group for each phone. These modes send one outage/recovery notification per group, not repeated email on every check or on every additional phone dropping while a group is down.

The wizard previews actual status before saving. Nonexistent IDs, NonQual or unknown reachability prevent saving: verify numbers and have the administrator review PJSIP qualify settings. NonQual means reachability is not established; this tool does not silently treat it as an outage. Known AORs with no contacts are unavailable and may be saved. Legacy chan_sip is not supported here.

OPTION 11 - PREVIEW EXTENSION STATUS

```
bash /root/freepbx-care.sh extensions-status
```

Option 11 / this command does not send email or restart Asterisk. Check that every configured extension has the expected available/unavailable status. Then use option 2 if the Care timer is not installed/enabled yet. The existing timer runs both checks; no separate cron job or legacy installer is needed.

Extension alerts go to the same recipient/sender set in option 1 and use the SMTP relay already configured. They NEVER trigger an Asterisk restart. Three consecutive unavailable checks are required by default, then one recovery notification when the mode's recovery condition is met. Failed local mail submissions are retried; accepted queue submissions still

need working SMTP. At most five extension notification submissions are attempted per scheduled check; additional pending group changes are handled by later checks.

Checks respect the same maintenance/backup pauses and boot/resume grace as PBX monitoring. An unhealthy PBX suppresses property checks, avoiding a flood of property alerts for a core PBX failure. Repeated failed/unknown PJSIP queries generate a separate 'Extension collection' alert instead of false phone outages. Use a controlled phone disconnection, with approval, to validate alerts; do not stop Asterisk to test an extension alert. Check SMTP logs and actual receipt.

EDIT OR DISABLE EXTENSION MONITORING LATER

Choose option 10. To edit groups, answer y to replacing groups and enter ALL groups you want to keep. Answer n to replacement to retain the displayed list. Answer n at the first enable prompt to disable extension checks while retaining the group settings. This leaves Asterisk health monitoring enabled. Groups are stored locally and preserved by normal updates. Choose stable unique names.

OPTION 12 - MARIADB / CRM SERVICE POLICY

Choose option 12, or run:

```
bash /root/freepbx-care.sh services
```

For this fleet (CRM not used), answer:

```
Automatically start stopped MariaDB? y/n [y]:
  y
Expect CRM FastAGI on 127.0.0.1:4573? y/n [n]:
  n
```

MariaDB automatic start also requires automatic recovery enabled in option 1. With option 1 set to n, all automatic starts/restarts remain disabled. After three failed checks by default, the watchdog may start a stopped, enabled local mariadb.service. It does not restart a running database on a failed ping, start disabled/masked units, or interfere with a pending service job. The same 30-minute cooldown and two-attempts-per-hour limits apply separately to database starts. A start timeout blocks further attempts until administrator review and resume. Pause monitoring before deliberately stopping MariaDB for maintenance.

While the database is unavailable, Asterisk restarts and dependent extension/CRM checks are deferred. A successful database start is followed by a full health recheck. This does not repair corruption, fix bad credentials, free disk space or solve memory exhaustion. Root socket ping proves service reachability, not application schema correctness. Other SQL errors still need investigation.

CRM n disables only the optional monitor check; it does NOT disable the CRM module itself. For unused CRM use the one-time procedure in section 8H. If a future site uses CRM, preserve its module and answer y to the listener check. It alerts after consecutive failures and reports listener recovery, but never disables modules, restarts Asterisk, or executes an AGI script. An open TCP port alone does not prove that CRM integration works; verify actual calls.

OPTION 13 - REVIEW AND DISABLE UNUSED OPTIONAL MODULES

Use this during approved maintenance with a current recoverable PBX backup. The helper changes ONE module per run. It does not guess whether business features are unused, uninstall modules, force dependencies, or request an Asterisk restart. A configuration reload and module hooks can still affect features/services, so test afterwards and do not use it during live calls.

Read-only inventory, with optional-feature explanations:

```
bash /root/freepbx-care.sh module-audit
```

Interactive review/disable (also menu 13):

```
bash /root/freepbx-care.sh modules
```

The helper displays installed/local modules and their versions/status. Only these optional features may be disabled here, when installed AND Enabled:

```
sangomacrm       CRM integration; confirmed unused in this fleet
zulu             Zulu desktop/mobile client
sangomaconnect   Sangoma Connect softphones
sangomartapi     Sangoma real-time API/integrations
restapps         Phone applications
ucp              User Control Panel and its web features
faxpro           Advanced fax features
fax              Fax support
xmpp             Chat/messaging integration
recording_report Call recording reporting interface
```

Other than CRM, these are REVIEW candidates, not a recommendation to disable. Ask the site which clients, phone apps, fax, web voicemail and integrations it uses. Dependencies between optional modules may block changes. Disabled does not mean uninstalled; disabled modules may still need security updates. No calls or low CPU usage do not prove a feature is unused. Disabling a module may not stop an already running process or materially reduce memory use.

Everything else is protected/audit-only here, including framework, core, PM2, firewall, user management, sysadmin, voicemail and routing/calling modules. Never stop all Node/PM2 processes or disable the firewall to free memory.

Example for the confirmed unused CRM integration:

```
Unused optional module to disable (Enter = audit only):
  sangomacrm
```

The script displays the exact proposed change and feature impact. To apply:

```
Type DISABLE sangomacrm to confirm this feature is unused and apply:
  DISABLE sangomacrm
```

Enter at the first prompt leaves modules unchanged. Any other confirmation cancels. If a dependency, unhealthy PBX, active calls, unknown channel count, unsupported output format or concurrent operation blocks the change, stop for review rather than bypassing it. Native reverse-dependency checks run before disable, and native disable also checks dependencies. Never add --force.

The helper saves inventory, monitor settings, logs and RESTORE.txt under:

```
/var/lib/freepbx-care/module-change-TIMESTAMP/
```

This is an audit/settings backup, NOT a full PBX/database backup. A successful operation verifies the selected module is Disabled and other inventory entries are unchanged, reloads configuration and checks MariaDB/CLI/AMI health. Disabling sangomacrm also turns off its optional FastAGI monitor check. Existing manual and backup pauses remain in place. Test calls and affected features afterwards.

If disable/reload/verification fails or times out, this file holds scheduled monitoring and automatic recovery across reboots until technician review:

```
/var/lib/freepbx-care/module-change-pending.json
```

Read the saved logs and RESTORE.txt; check whether fwconsole work is still running. Do not blindly retry or delete the marker. Native module hooks are not a transaction, so the helper does not claim a complete automatic rollback. If restoring the selected module is appropriate, RESTORE.txt gives the exact enable/reload commands. Restore PBX health and review module state, then run:

```
bash /root/freepbx-care.sh resume
```

Resume refuses to clear a module hold while fwconsole is running or PBX health checks fail. It also clears a manual pause and recovery latches; coordinate with maintenance. A separate backup pause remains independent.

OPTION 14 - INVESTIGATE CRASH/RESOURCE ISSUES

Run during/soon after a failure, before rebooting or deleting evidence:

```
bash /root/freepbx-care.sh crash-audit
```

This prints a cause checklist and saves a root-only diagnostic snapshot under /var/log/freepbx-care/. It checks memory, disk space/inodes, service failures, kernel logs, process names, socket listeners and available container resource limits/pressure counters. Collection has a 45-second budget and may skip later commands; inaccessible host/kernel counters are recorded as errors. No services/modules are changed. These expanded diagnostics are on demand or incident-only, not extra probes on every normal scheduled check. Share only reviewed/redacted evidence with the administrator; do not upload it publicly.

5. SMTP PROMPT ANSWERS

Choose menu option 3, or run directly:

```
bash /root/freepbx-care.sh smtp
```

For the current company support@famepbx.com relay, enter:

```
SMTP relay DNS hostname (must match its certificate):
  premium117-1.web-hosting.com
SMTP port [587]:
  465
TLS mode: starttls or implicit (465) [implicit]:
  implicit
SMTP username:
  support@famepbx.com
SMTP password:
  Actual mailbox/app password from the secure credential store.
  Input is hidden; this is normal. Do not put it in a shell command.
Monitor sender email:
  support@famepbx.com
```

These settings were verified for this mailbox. If the provider migrates the account, use its updated SMTP hostname and matching updated instructions. The email domain alone does not determine the relay hostname.

Do NOT use 162.0.229.110 at the hostname prompt: it produced a certificate 'IP address mismatch'. Keep certificate verification enabled and use the DNS hostname above. The older SMTP installer could deliver using an IP because its TLS settings did not require the same certificate identity checks.

For another approved provider use its documented SMTP hostname, credentials and port. Port 587 normally uses starttls; port 465 normally uses implicit. Do not reuse this account's settings for another provider.

Setup may install missing mail packages and restarts Postfix, not Asterisk. Relevant backups: /var/lib/freepbx-care/smtp-backup-TIMESTAMP/. Unrelated Postfix configuration is preserved; existing sender rewriting can still affect voicemail. Test actual delivery and sender identity after changes.

6. VERIFY EMAIL DELIVERY

```
bash /root/freepbx-care.sh test-mail
postqueue -p
tail -n 60 /var/log/mail.log
```

Match the new message's time and queue ID. 'status=sent' with a successful relay response means relay acceptance; confirm the recipient's inbox/spam folder too. An empty queue alone does not prove inbox delivery. If mail.log is unavailable, try:

```
journalctl --since "15 minutes ago" --no-pager | grep -i postfix
```

Certificate error / IP mismatch: rerun section 5 using the DNS hostname. Old errors remain in logs, so compare timestamps with your new attempt. Authentication failure / 535: verify the mailbox/app password with the account administrator and rerun SMTP setup. Never print/share the password map. Connection timeout/refused: administrator checks DNS, outbound port 465 and provider availability from this PBX. Deferred after intentional reset: finish option 3 using the current main script; do not manually remove /etc/freepbx-care/smtp-reset.json.

After fixing the cause, retry a specific queued message:

```
postqueue -i QUEUE_ID
```

Replace QUEUE_ID with the actual ID from postqueue -p, then recheck logs and receipt. Do not type the placeholder literally or delete the queue.

7. ROUTINE OPERATION, MAINTENANCE AND UPDATES

Status and diagnostics without deliberately requesting recovery:

```
bash /root/freepbx-care.sh status
bash /root/freepbx-care.sh diagnose
```

Diagnostics go under /var/log/freepbx-care/ and may contain sensitive call metadata. The 'check' command can trigger recovery; it is not a read-only test.

Before intentionally stopping/restarting Asterisk, restoring, backing up or performing disruptive maintenance, pause for enough time:

```
bash /root/freepbx-care.sh pause --minutes 120
```

Adjust 120 to the planned duration and extend if work overruns. Pauses expire automatically and also suppress monitoring/alerts. Wait for an existing check or recovery to finish before disruptive work.

After maintenance, check Asterisk is healthy, then:

```
bash /root/freepbx-care.sh resume
bash /root/freepbx-care.sh status
```

A separate backup pause remains independent of manual resume. Only after the backup finishes, if status shows such a pause, use:

```
bash /root/freepbx-care.sh backup-resume
```

This leaves a three-minute settling period. Do not clear an active backup's pause.

Disable future scheduled checks:

```
bash /root/freepbx-care.sh disable
```

This does not cancel a recovery already running. Enable again with:

```
bash /root/freepbx-care.sh install
```

Change recipient/recovery settings: menu option 1. Change SMTP: option 3, then option 5 and confirm receipt. Change/disable extension alerts: option 10; preview with option 11. Change MariaDB/CRM monitoring policy: option 12. Review unused modules: option 13. Investigate resource crashes: option 14. Saved settings apply to later checks; no legacy cleanup is needed.

UPDATE AN EXISTING WORKING FREEPBX CARE INSTALLATION

Update and verify one server first. Keep its settings. Back up the current script:

```
cp -p /root/freepbx-care.sh /root/freepbx-care.before-update-$(date +%Y%m%d-%H%M%S).sh
```

Pause as above, then download/replace ONLY the main script using section 2.

```
bash /root/freepbx-care.sh --version
bash /root/freepbx-care.sh preflight
bash /root/freepbx-care.sh install
bash /root/freepbx-care.sh test-mail
```

Confirm receipt and healthy status, resume, and verify the timer as in section 4. An existing pause remains until expiry or resume. Do not rerun cleanup or SMTP just to update the script. If any step fails, stop and obtain administrator review before resuming. Keep the previous script for reviewed rollback.

8. MIGRATION AND INSTALLATION TROUBLESHOOTING

A. 'A monitor/recovery is running; wait for it to finish and retry'

Inspect without stopping Asterisk:

```
lslocks -o PID,COMMAND,PATH
fuser -v /tmp/ast_mon.lock /run/freepbx-care.lock
tail -n 12 /var/log/asterisk_monitor.log
```

Wait for a real monitor, fwconsole or recovery operation to finish. Do not kill safe_asterisk/asterisk or delete the lock to force cleanup.

Some old monitors left /tmp/ast_mon.lock inherited by running Asterisk daemons. When safe_asterisk/asterisk are its ONLY holders and the Asterisk CLI responds, the current cleaner provides a guarded migration:

```
bash /root/cleanup-legacy-freepbx.sh --apply --retire-inherited-lock
```

It verifies holders/readiness, backs up and disables recognized legacy cron, retires the old lock and cleans up without signalling Asterisk. It refuses other holders or uncertain conditions. If refused, collect the error and ask the administrator. This does not retire /run/freepbx-care.lock. Backups: /var/lib/freepbx-care/lock-retirement-TIMESTAMP/. If retirement

fails after disabling old cron, jobs deliberately remain disabled to prevent a race. Finish migration promptly with administrator help. After success, continue section 4.

B. 'Legacy monitor jobs remain'

Run section 3 preview/apply. Do not rerun the old installer or bypass the guard. Custom or chained jobs need administrator review.

C. Unsupported platform or Debian repository errors

Confirm Debian 12 / FreePBX 17 and the current script. If apt reports stable changing to trixie, or the script flags moving suites, stop for repository review. Debian 12 needs appropriate bookworm suites. Do not blindly accept a release change, upgrade Debian, or replace every 'stable' string: third-party repositories can legitimately use different suite names.

D. CLI unhealthy, AMI config_error/auth_error or preflight failed

```
bash /root/freepbx-care.sh status
bash /root/freepbx-care.sh diagnose
systemctl status mariadb --no-pager
```

Have the administrator fix the local AMI/FreePBX/database issue before enabling automatic recovery. Do not publish /etc/freepbx.conf, guess credentials, use the SMTP password for AMI, or expose AMI port 5038 to the Internet.

E. Recovery timeout / recovery_blocked

Further automatic attempts are blocked after a recovery timeout. Review diagnostics and /var/log/freepbx-care/last-restart.log. The administrator must check any remaining fwconsole operation and the PBX's actual condition first. Only after review, 'resume' clears the timeout latch. Do not repeatedly resume to force retries or run fwconsole restart while calls may be active. For database_recovery_blocked, also review last-database-start.log in the same directory and systemctl status mariadb. A systemd start job may still be running after a client timeout. Resume clears both latches after review.

F. Cleanup or SMTP failed partway through

Read the complete error and backup path. Rollback is attempted; do not assume everything was restored. The Care timer may remain disabled after failed cleanup. Verify health, mail and timer state, then resolve the failure and finish setup. Preserve backups. Do not restore all old cron blindly and create competing monitors.

G. SQLSTATE[HY000] [2002] Connection refused

This points to an unavailable database connection; it does not prove why the service stopped. If local MariaDB is stopped and supposed to run:

```
systemctl status mariadb --no-pager -l
systemctl start mariadb
mariadb-admin --protocol=socket --connect-timeout=3 ping
bash /root/freepbx-care.sh status
```

If it is disabled or masked, review why before enabling/unmasking it. Do not restart a running database blindly. Collect evidence if start fails:

```
journalctl -u mariadb -n 100 --no-pager
journalctl -k --since "24 hours ago" --no-pager | grep -Ei 'oom|out of memory|killed process'
df -h / /var/lib/mysql
free -h
```

Kernel logs may require host-side inspection for a container. Investigate OOM, disk/full filesystem, permissions or database errors. A successful start alone does not explain the outage. Never remove database files to force a start.

H. FastAGI sangomacrm.agi: 127.0.0.1:4573 connection refused

This is a separate local CRM listener failure; Asterisk CLI/AMI can stay healthy. For this fleet CRM is not used. Disable the unused module ONCE in an approved maintenance window, after reviewing module dependencies:

```
bash /root/freepbx-care.sh pause --minutes 60
fwconsole ma list
```

If sangomacrm is installed/enabled and confirmed unused, run:

```
fwconsole ma disable sangomacrm
```

Only after disable succeeds, apply the configuration:

```
fwconsole reload
```

If dependencies block the change, stop for administrator review; do not force disable dependent modules. Alternatively use Admin > Module Admin to disable Sangoma CRM, then Apply Config. Test inbound/outbound/hangup paths and confirm NEW sangomacrm FastAGI warnings stop, then resume monitoring:

```
bash /root/freepbx-care.sh resume
```

In option 12 leave the CRM listener check at n. Do not reinstall CRM merely to satisfy a port check, or disable all Node/PM2 services. Disabling CRM removes its integration features; do not apply this procedure to a site that uses them.

If a site USES CRM, preserve it and investigate its listener/process instead:

```
ss -lntp | grep ':4573'
timeout 20 fwconsole pm2 --list
tail -n 100 /var/log/asterisk/full
```

Port refusal does not identify the failed component by itself. Have the administrator review CRM/PM2 logs and repair the particular service; no blanket Asterisk restart or automatic module disable is performed by the watchdog.

I. OTHER CRASH/STALL CAUSES TO INVESTIGATE

Memory exhaustion/OOM: compare kernel logs and changes in memory.events with the incident time. A 1 GB server has limited headroom for MariaDB, Apache and Node/PM2 services. Historical swap/OOM counts alone do not diagnose a current failure. Fix capacity/leaks or unused services after identifying the cause.

Full disks OR inodes: recordings, voicemail, CDRs, backups and runaway logs can exhaust space or file entries. Use df -h and df -i. Set reviewed retention and backup policies; do not wildcard-delete recordings, logs, database or state files.

CPU quota, process limits and I/O stalls: check cgroup cpu.stat/cpu.max, pressure counters, pids.events and storage/host health. Busy recordings, transcoding, conferences or overlapping backups can add load. Container-wide or host pressure may exist even if one process snapshot looks quiet.

Other services and configuration: inspect failed services, MariaDB/Apache/PM2 logs, recent module changes and stuck update/backup tasks. Permission errors, bad database options or damaged storage need specific repairs, not repeated restarts. Preserve evidence and test a recovered server before wider changes.

Conflicting jobs: old cron monitors, another recovery tool or a maintenance task may restart services unexpectedly. Review recognized legacy jobs and coordinate a single recovery owner. Pause before planned disruptive work.

Updates and security: review supported Debian suites, module updates and vendor advisories. Test updates before broad deployment. Investigate SIP authentication storms or unusual traffic with the administrator; preserve firewall/Fail2ban protection. Never expose AMI or disable security to troubleshoot.

DNS, trunks, NAT, certificates or phone/network faults can break calls or mail without crashing Asterisk. Use call traces and provider/network diagnostics; a healthy CLI/AMI check alone does not prove end-to-end call service.

9. MONITOR BEHAVIOR AND HANDOFF NOTES

The timer starts five minutes after boot, then schedules checks about 60 seconds after the previous check finishes, plus up to 15 seconds of randomized delay. Long checks, cooldowns and settling periods delay recovery; three failures do not guarantee a restart in three minutes.

Standard settings: three consecutive failed checks, 30-minute cooldown, maximum two recovery attempts per hour. CLI and AMI are evaluated together. AMI authentication/configuration errors alone do not force a restart of an otherwise responsive Asterisk. Active-channel deferral helps when CLI responds; it cannot guarantee call preservation when CLI is unresponsive. Recovery can interrupt calls and does not fix the underlying cause of repeated crashes/timeouts.

Optional PJSIP reachability polling runs only after local technician setup. It observes Asterisk's contact status, not end-to-end call quality; an Avail contact does not prove a successful phone call. Use external monitoring to detect a powered-off or unreachable whole server.

Operational files (keep credentials and diagnostics private):

```
/root/freepbx-care.sh                  Administrator script
/etc/freepbx-care/config.json          Settings; may contain secrets
/etc/postfix/freepbx-care-sasl          SMTP credentials, plus its .db file
/var/lib/freepbx-care/                 State, pauses and setup backups
/var/lib/freepbx-care/extensions.json   Persistent extension alert/debounce state
/var/lib/freepbx-care/module-change-*/  Private module audit and restore plans
/var/log/freepbx-care/                 Diagnostics and recovery output
```

10. GITHUB FILES TO PUBLISH - MAINTAINER

Keep these at the ROOT of workimirp201/freepbx-script, branch main:

```
freepbx-care.sh                Current reviewed main script
cleanup-legacy-freepbx.sh      Current standalone cleaner with lock fix
STANDALONE-INSTRUCTIONS.txt    This colleague instruction file
```

The first two are required by the download commands. If the standalone cleaner has not been uploaded, add it too. No new standalone installer wrapper is needed. The ZIP and SHA256SUMS may also be published as distribution artifacts.

Publish reviewed sources and placeholder examples only. Never upload passwords, actual config.json, credential maps, real SMTP profiles, private keys, server backups or diagnostic logs. Update this guide if prompts/relay settings change, and check the raw download URLs after publishing.